

PROBLEM SET 1

1. a)

i) $J(\theta) = \frac{1}{2} \beta \theta^2 \Rightarrow J'(\theta) = \frac{1}{2} \beta 2\theta = \beta \theta \Rightarrow \theta^{[t]} = \theta^{[t-1]} - \alpha \underbrace{\beta \cdot \theta^{[t-1]}}_{J'(\theta^{[t-1]})} = \theta^{[t-1]} (1 - \alpha \beta) = (1 - \alpha \beta) \theta^{[t-2]} - \alpha \underbrace{\beta \theta^{[t-2]}}_{J'(\theta^{[t-2]})} = (1 - \alpha \beta)^2 \theta^{[t-2]} = \dots = (1 - \alpha \beta)^t \theta^{[0]}$

So $\theta^{[t]} = (1 - \alpha \beta)^t \cdot \theta^{[0]} \quad t \in \mathbb{N}_{\geq 1}, \theta^{[0]} \neq 0$

For $\theta^{[t]}$ to converge to a scalar, $(1 - \alpha \beta)^t$ has to be bounded when t tends to ∞ , which is only possible if $|1 - \alpha \beta| \leq 1$ with $1 - \alpha \beta \neq -1$ ($\lim_{t \rightarrow \infty} (-1)^t \theta^{[0]}$ does not converge)

If $\alpha = 0$, $\theta^{[t]}$ is not updated and GD cannot run properly. So let us suppose $\alpha \neq 0$. In other words, $1 - \alpha \beta \neq -1$

Therefore, $-1 < 1 - \alpha \beta < 1 \Leftrightarrow \begin{cases} -1 < 1 - \alpha \beta \Leftrightarrow -2 < -\alpha \beta \Leftrightarrow \alpha < 2\beta^{-1} \\ 1 - \alpha \beta < 1 \Leftrightarrow -\alpha \beta < 0 \Leftrightarrow \alpha > 0 \end{cases} \Leftrightarrow \alpha \in (0, 2\beta^{-1})$ is necessary for $\theta^{[t]}$ to converge to a scalar θ^* .

$\lim_{t \rightarrow \infty} \theta^{[t]} = 0 \cdot \theta^{[0]} = 0 = \theta^*$. let us see that $\theta^* = \arg \min_{\theta} J(\theta)$.

Since $J(\theta)$ is a convex parabola, it has one and only one minimum, which is found at its vertex or when $J'(\theta) = 0 \Leftrightarrow \beta \theta = 0 \Leftrightarrow \theta = 0 = \arg \min_{\theta} J(\theta) = \theta^* = 0$

ii) $|\theta^{[T]} - 0| = |(1 - \alpha \beta)^T \theta^{[0]}| = |1 - \alpha \beta|^T \cdot |\theta^{[0]}| \leq \frac{\epsilon}{|\theta^{[0]}|} \cdot |\theta^{[0]}| = \epsilon \quad \forall \epsilon > 0$. So given $\epsilon > 0$, $T = \max \{0, \frac{\ln(\epsilon/|\theta^{[0]}|)}{\ln|1 - \alpha \beta|}\}$ for $|\theta^{[T]} - \theta^*| = |\theta^{[T]}| \leq \epsilon$

$\theta^* = \arg \min_{\theta} J(\theta) = 0$

$T = \max \{0, \frac{\log(\epsilon/|\theta^{[0]}|)}{\log|1 - \alpha \beta|}\}$

$|1 - \alpha \beta|^T \leq \frac{\epsilon}{|\theta^{[0]}|} \Leftrightarrow T \cdot \log|1 - \alpha \beta| \leq \log\left(\frac{\epsilon}{|\theta^{[0]}|}\right) \Leftrightarrow T \geq \frac{\log(\epsilon/|\theta^{[0]}|)}{\log|1 - \alpha \beta|}$

$|1 - \alpha \beta| < 1, \alpha \in (0, 2\beta^{-1})$ from a) i)

We conclude that as α approaches 0 and $2\beta^{-1}$, T becomes extremely large and convergence becomes very slow (due to very small step size or 'bouncing around' the minimum with too large or a step size respectively).

As α approaches β^{-1} , T becomes very small and approaching fastest convergence.

b) $J(\theta) = \frac{1}{2} \sum_{i=1}^d \beta_i \theta_i^2 \Rightarrow \nabla J(\theta) = (\beta_1 \theta_1, \dots, \beta_d \theta_d)^T \Rightarrow \theta^{[t]} = \theta^{[t-1]} - \alpha \begin{pmatrix} \beta_1 \theta_1^{[t-1]} \\ \vdots \\ \beta_d \theta_d^{[t-1]} \end{pmatrix} = \begin{pmatrix} \theta_1^{[t-1]} (1 - \alpha \beta_1) \\ \vdots \\ \theta_d^{[t-1]} (1 - \alpha \beta_d) \end{pmatrix} = \dots = \begin{pmatrix} \theta_1^{[0]} (1 - \alpha \beta_1)^t \\ \vdots \\ \theta_d^{[0]} (1 - \alpha \beta_d)^t \end{pmatrix}$

For $\theta^{[t]}$ to converge to a vector $\theta^* \in \mathbb{R}^d$, $|1 - \alpha \beta_i| < 1 \quad \forall i \in \{1, \dots, d\}$ is necessary

Which means that $\alpha \in (0, 2 \cdot \min_{i=1, \dots, d} \beta_i^{-1})$, derived trivially from a) i)

$\theta^* = \lim_{t \rightarrow \infty} \theta^{[t]} = \vec{0}$

$|1 - \alpha \beta_i| < 1 \quad \forall i \in \{1, \dots, d\}$

c) i) On GitHub

ii) Discussion: it seems like the trajectories of have somewhat rotated with the rotation of the matrix. In both cases, a big enough learning rate results in GD not converging

d) $J(\theta)$ bounded from below $\Rightarrow \exists m \in \mathbb{R}$ s.t. $m \leq J(\theta) \quad \forall \theta \in \mathbb{R}^d$

So if we show that $J(\theta^{[t]}) < J(\theta^{[t-1]}) \quad \forall t \in \mathbb{N}$, when $t \rightarrow \infty$, $J(\theta^{[t]}) \in [m, J(\theta^{[0]})] \Rightarrow J(\theta^{[t]})$ converges to some finite value as $t \rightarrow \infty$

$J(\theta^{[t]}) = J(\theta^{[t-1]}) + \nabla J(\theta^{[t-1]})^T (\theta^{[t]} - \theta^{[t-1]}) + \frac{1}{2} (\theta^{[t]} - \theta^{[t-1]})^T \nabla^2 J(\theta^{[t-1]}) (\theta^{[t]} - \theta^{[t-1]}) + o(\|\theta^{[t]} - \theta^{[t-1]}\|^2)$. Let's prove that $J(\theta^{[t]}) - J(\theta^{[t-1]}) < 0$

Taylor's Theorem
for some $0 \leq \zeta < 1$

J convex $\Rightarrow H = \nabla^2 J(\theta)$ is symmetric and positive semi-definite $\Rightarrow \text{sp}(H) \subset \mathbb{R}_{\geq 0}$

H symmetric $\Rightarrow \exists P$ invertible s.t. $H = P^T D P$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $\text{sp}(H) = \{\lambda_1, \dots, \lambda_n\}$. P being orthogonal matrix ($\Rightarrow P^{-1} = P^T$)

Moreover, $\theta^{[t]} = \theta^{[t-1]} - \alpha \nabla J(\theta^{[t-1]}) \Leftrightarrow \nabla J(\theta^{[t-1]}) = -\frac{1}{\alpha} (\theta^{[t]} - \theta^{[t-1]})$. Let $v^{[t]} = \theta^{[t]} - \theta^{[t-1]}$

Therefore, $J(\theta^{[t+1]}) - J(\theta^{[t-1]}) = -\frac{1}{\alpha} \|v^{[t]}\|_2^2 + \frac{1}{2} (v^{[t]})^T P^T \Delta P v^{[t]} \underset{0 < \alpha < \frac{1}{\beta_{\max}}}{<} -\beta_{\max} \|v^{[t]}\|_2^2 + \frac{1}{2} (P v^{[t]})^T \Delta (P v^{[t]}) = -\beta_{\max} \|v^{[t]}\|_2^2 + \frac{1}{2} (u^{[t]})^T \Delta u^{[t]} <$

$$< -\beta_{\max} \|v^{[t]}\|_2^2 + \frac{\beta_{\max}}{2} \|u^{[t]}\|_2^2 = \beta_{\max} \left(\frac{1}{2} \|u^{[t]}\|_2^2 - \|v^{[t]}\|_2^2 \right) = -\frac{\beta_{\max}}{2} \|v^{[t]}\|_2^2 < 0$$

$$\|u^{[t]}\|_2^2 = \langle u^{[t]}, u^{[t]} \rangle = (P v^{[t]})^T P v^{[t]} = (v^{[t]})^T P^T P v^{[t]} = (v^{[t]})^T v^{[t]} = \|v^{[t]}\|_2^2$$

$\downarrow$
 $P^T = P^{-1}$

Optional part: in d) we did not technically get the inequality we were looking for. However, upon closer inspection...

$$\nabla J(\theta^{[t-1]})^T (\theta^{[t]} - \theta^{[t-1]}) = \nabla J(\theta^{[t-1]})^T (\theta^{[t-1]} - \alpha \nabla J(\theta^{[t-1]}) - \theta^{[t-1]}) = -\alpha \|\nabla J(\theta^{[t-1]})\|_2^2$$

$$\text{So } J(\theta^{[t]}) < J(\theta^{[t-1]}) - \alpha \|\nabla J(\theta^{[t-1]})\|_2^2 + \frac{\beta_{\max}}{2} \|\theta^{[t]} - \theta^{[t-1]}\|_2^2 = J(\theta^{[t-1]}) - \alpha \|\nabla J(\theta^{[t-1]})\|_2^2 + \frac{\alpha^2 \beta_{\max}}{2} \|\nabla J(\theta^{[t-1]})\|_2^2$$

Now, $\|\nabla J(\theta^{[t-1]})\|_2^2 < \frac{1}{\alpha} [J(\theta^{[t-1]}) - J(\theta^{[t]})] + \|\nabla J(\theta^{[t-1]})\|_2^2 \left(\frac{\alpha \beta_{\max}}{2} - 1 \right) < \frac{1}{\alpha} [J(\theta^{[t-1]}) - J(\theta^{[t]})] \xrightarrow{t \rightarrow \infty} 0$ since $\lim_{t \rightarrow \infty} J(\theta^{[t]}) = L = \lim_{t \rightarrow \infty} J(\theta^{[t-1]})$

$\downarrow$
 $\alpha < \frac{1}{\beta_{\max}} \Leftrightarrow \beta_{\max} < \frac{1}{\alpha} \Rightarrow$
 $\Rightarrow \frac{\alpha \beta_{\max}}{2} - 1 < \frac{1}{2} - 1 < 0$

e) $J(\theta) > 0$ (bounded from below) and $J(\theta)$ clearly twice differentiable. Indeed,

$$J(\theta) = \frac{1}{2} \left\| \begin{pmatrix} \sum_i X_{1i} \theta_i - y_1 \\ \vdots \\ \sum_i X_{ni} \theta_i - y_n \end{pmatrix} \right\|_2^2 = \frac{1}{2} \sum_k \left(\sum_i X_{ki} \theta_i - y_k \right)^2 \Rightarrow \frac{\partial}{\partial \theta_i} J(\theta) = \frac{1}{2} \sum_k \left[2 \cdot \left(\sum_i X_{ki} \theta_i - y_k \right) \cdot \frac{\partial}{\partial \theta_i} \left(\sum_i X_{ki} \theta_i - y_k \right) \right] = \sum_k \sum_i (X_{ki} \theta_i - y_k) X_{ki} \Rightarrow$$

$$\Rightarrow \frac{\partial}{\partial \theta_i} \frac{\partial}{\partial \theta_j} J(\theta) = \sum_k \sum_i \frac{\partial}{\partial \theta_j} [(X_{ki} \theta_i - y_k) X_{ki}] = \sum_k X_{kj} X_{ki} = \sum_k X_{ik}^T X_{ki} \Rightarrow \nabla^2 J(\theta) = X^T X, \text{ whose largest eigenvalue is less than } \beta_{\max} \text{ for any } \theta$$

Therefore, from d): $J(\theta^{[t]}) < J(\theta^{[t-1]}) - \alpha \|\nabla J(\theta^{[t-1]})\|_2^2 \left(1 - \frac{\alpha \beta_{\max}}{2} \right) < \infty$ with the assumption that $\alpha \in (0, \frac{1}{\beta_{\max}})$

$\downarrow t \rightarrow \infty$ $\downarrow t \rightarrow \infty$
 $L \in [m, J(\theta^{[0]})]$ 0 from the optional part of d)
 from d)

2. a)

i) Suppose our training set is of size n . From lectures, $X = \begin{pmatrix} (x^{(1)})^T \\ \vdots \\ (x^{(n)})^T \end{pmatrix} \in \mathbb{R}^{n \times d}$, $y = \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{pmatrix} \in \mathbb{R}^n$, $\theta \in \mathbb{R}^d$

$$X\theta - y = \begin{pmatrix} \theta^T x^{(1)} - y^{(1)} \\ \vdots \\ \theta^T x^{(n)} - y^{(n)} \end{pmatrix}$$

Let $W = \begin{pmatrix} 2w^{(1)} & 0 \\ 0 & 2w^{(n)} \end{pmatrix}$. Then, $W(X\theta - y) = \begin{pmatrix} 2w^{(1)}(\theta^T x^{(1)} - y^{(1)}) \\ \vdots \\ 2w^{(n)}(\theta^T x^{(n)} - y^{(n)}) \end{pmatrix}$ and $(X\theta - y)^T W (X\theta - y) = 2 \sum_{i=1}^n w^{(i)} (\theta^T x^{(i)} - y^{(i)})^2$

So $J(\theta) = (X\theta - y)^T W (X\theta - y)$ with $W = 2 \text{diag}(w^{(1)}, \dots, w^{(n)})$

ii) $\nabla_{\theta} J(\theta) = \nabla_{\theta} ((X\theta - y)^T W (X\theta - y)) = \nabla_{\theta} ((X\theta)^T W X\theta - (X\theta)^T W y - y^T W X\theta + y^T W y) = \nabla_{\theta} (\theta^T X^T W X\theta - \underbrace{y^T W X\theta}_{\text{Since } W \text{ is diagonal, } X^T W X \text{ is symmetric}} - \underbrace{y^T W X\theta}_{\text{Since } W \text{ is diagonal, } X^T W X \text{ is symmetric}}) =$
 $= 2X^T W X\theta - 2X^T W y$

$\nabla_{\theta} J(\theta) = 0 \iff [\theta = (X^T W X)^{-1} X^T W y = \arg \min_{\theta} \nabla_{\theta} J(\theta)]$

iii) $L(\theta) = p(y|x; \theta) = \prod_{i=1}^n p(y^{(i)}|x^{(i)}; \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma^{(i)}} \exp\left\{-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2}\right\}$

$\ell(\theta) = \sum_{i=1}^n \log\left(\frac{1}{\sqrt{2\pi}\sigma^{(i)}} \exp\left\{-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2}\right\}\right) = \underbrace{\sum_{i=1}^n \log\left(\frac{1}{\sqrt{2\pi}\sigma^{(i)}}\right)}_{\text{Constant (doesn't depend on } \theta)} - \sum_{i=1}^n \frac{(y^{(i)} - \theta^T x^{(i)})^2}{2(\sigma^{(i)})^2}$
 $\rightarrow \sum_{i=1}^n w^{(i)} (y^{(i)} - \theta^T x^{(i)})^2$ where $w^{(i)} = \frac{1}{2(\sigma^{(i)})^2}$. It is a LWR problem

Since $\nabla \ell(\theta)$ is only concerned with the second term, the LWR problem, solving it gives us the max. likelihood estimate for θ

b) On GitHub. It seems to be underfitting. The model isn't expressive enough to capture the sine/cosine-like pattern to the left and right of the central "M" or the τ -value isn't the best.

c) On GitHub.

3.

a) $J(\theta) = \frac{1}{2} \sum_{i=1}^n \|y^{(i)} - \theta^T \hat{x}^{(i)}\|_2^2$ with $\{(\hat{x}^{(i)}, y^{(i)})\}_{i=1}^n$ training set

BGD update rule: $\theta^{[t]} = \theta^{[t-1]} - \alpha \nabla_{\theta} J(\theta^{[t-1]})$ where $t \in \mathbb{N}$ is the iteration and $\alpha \in \mathbb{R}$ the step size

$$J(\theta) = \frac{1}{2} (\hat{X}\theta - y)^T (\hat{X}\theta - y) \Rightarrow \nabla_{\theta} J(\theta) = \frac{1}{2} \nabla_{\theta} ((\hat{X}\theta)^T \hat{X}\theta - (\hat{X}\theta)^T y - y^T \hat{X}\theta + y^T y) = \frac{1}{2} (2\hat{X}^T \hat{X}\theta - 2\hat{X}^T y) = \hat{X}^T (\hat{X}\theta - y)$$

b) On Github.

c) On Github. As k increases, the model becomes more expressive and captures the oscillatory pattern of the data.

However, at $k=20$, the model seems to start following individual data points, which suggests increasing variance and potential overfitting (capturing noise).

d) On Github. New feature map greatly improves fitting for the lower k 's. In this case, up to $k=10$, the different model's predictions are almost superimposed. However, at $k=20$, we find that the model falls into the same issue as in the previous sub-question. The polynomial finally "wins" over the sine part of the model.

e) On Github. As the remark points out, the model is very clearly overfitting as k increases. At lower k 's ($k=1, 2$) it underfits.

4.

a) $\mathcal{J}(X^T(XX^T)^{-1}\vec{y} + \vec{c}) = \frac{1}{2n} \overbrace{\|XX^T(XX^T)^{-1}\vec{y}\|_2^2}^I + \|X\vec{c} - \vec{y}\|_2^2 = \|X\vec{c}\|_2^2 = \|\vec{0}\|_2^2 = 0$
 $\vec{c}^T X^T = 0 \quad \forall 1 \leq i \leq n$

Now, if $\mathcal{J}(\beta) = 0$, $\beta = X^T(XX^T)^{-1}\vec{y} + \vec{c}$?

$\mathcal{J}(\beta) = 0 \Leftrightarrow \|X\beta - \vec{y}\|_2^2 = 0 \Leftrightarrow X\beta - \vec{y} = \vec{0} \Leftrightarrow X\beta = \vec{y}$, which is equivalent to $X(\beta + \vec{c}) = \vec{y}$ where $\vec{c} \in \mathbb{R}^d$ in X 's nullspace

So, $X^T(XX^T)^{-1}X(\beta + \vec{c}) = X^T(XX^T)^{-1}\vec{y} \Leftrightarrow \beta = X^T(XX^T)^{-1}\vec{y} + \vec{c}$

b) $\|\beta\|_2^2 = \|X^T(XX^T)^{-1}\vec{y} + \vec{c}\|_2^2 = (X^T(XX^T)^{-1}\vec{y} + \vec{c})^T (X^T(XX^T)^{-1}\vec{y} + \vec{c}) = \|\rho\|_2^2 - 2\rho^T \vec{c} + \|\vec{c}\|_2^2 = \|\rho\|_2^2 + \|\vec{c}\|_2^2$

0 because $\rho^T = \vec{y}^T(XX^T)^{-1}X$ and $\vec{c}$ lies in the nullspace of X

So $\|\rho\|_2^2 = \|\beta\|_2^2 - \|\vec{c}\|_2^2 \leq \|\beta\|_2^2 \Rightarrow \|\rho\|_2 \leq \|\beta\|_2$ because $f(x) = \sqrt{x}$ is strictly increasing $\forall x \geq 0$ ($f'(x) = \frac{1}{2}(x)^{-1/2} = \frac{1}{2\sqrt{x}} > 0 \quad \forall x > 0$ and $\lim_{x \rightarrow 0^+} f(x) = +\infty$)

c) On Github.

d) $\beta^{(0)} = \vec{0} = X^T \vec{0}$

Induction step: $\beta^{[t+1]} = X^T \vec{v}_{t+1}$ for $\vec{v}_{t+1} \in \mathbb{R}^n \Rightarrow \beta^{[t+1]} = X^T \vec{v}_{t+1} - \frac{\eta}{n} X^T (XX^T \vec{v}_{t+1} - \vec{y}) = X^T [\vec{v}_{t+1} - \frac{\eta}{n} (XX^T \vec{v}_{t+1} - \vec{y})]$

So $\hat{\beta} = X^T w$ for some $w \in \mathbb{R}^n$

Therefore, since $\mathcal{J}(\hat{\beta}) = \frac{1}{2n} \|XX^T w - \vec{y}\|_2^2 = 0 \Leftrightarrow XX^T w - \vec{y} = \vec{0} \xrightarrow{XX^T \text{ invertible}} w = (XX^T)^{-1} \vec{y}$, we have that $\hat{\beta} = X^T(XX^T)^{-1} \vec{y}$

e) Let $\beta = \theta^{\text{oz}} - \phi^{\text{oz}}$. Then, $\mathcal{J}(\theta, \phi) = \frac{1}{4n} \sum_{i=1}^n (\theta_0 \beta (x^{(i)}) - y^{(i)})^2 = \frac{1}{4n} \|X\beta - y\|_2^2 = 0 \Leftrightarrow X\beta = y$

From a), $\beta = X^T(XX^T)^{-1} \vec{y} + \vec{c}$ where $\vec{c} \in \text{Ker}(X)$

f) On Github. The 3 model initializations fit the training well, achieving low training set error. However, $\text{init.} = 0.01$ achieves by far the best validation error.

g) On Github. The smaller the batch size, the lower validation error we get and our model generalizes better. Compared to $\text{init.} = 0.1$, we get better generalization through SGD on batch sizes 1 and 5. Moreover, it is worth noting that f) learning curves are more stable since SGD updates the parameter locally as opposed to GD which takes in the whole training set to update the parameters. These curves also converge quicker. Finally, as the batch size increases, the more stable the curve. However, it doesn't lead to better generalization.

5.

a) From previous exercises, $\nabla_{\beta} J_{\lambda}(\beta) = X^T(X\beta - \vec{y})$

Since J_{λ} is convex, by equalling $\nabla_{\beta} J_{\lambda}$ to $\vec{0}$ we find the global minimum. $\nabla J_{\lambda}(\beta) = 0 \Leftrightarrow X^T(X\beta - \vec{y}) + \lambda\beta = 0 \Leftrightarrow X^T\vec{y} = -\lambda\beta - X^T X\beta \Leftrightarrow \beta = (X^T X + \lambda I_{\text{dim}})^{-1} X^T \vec{y}$

b) On Github.

c) On Github.